Persistent Homology of Music Network with Three Different Distances

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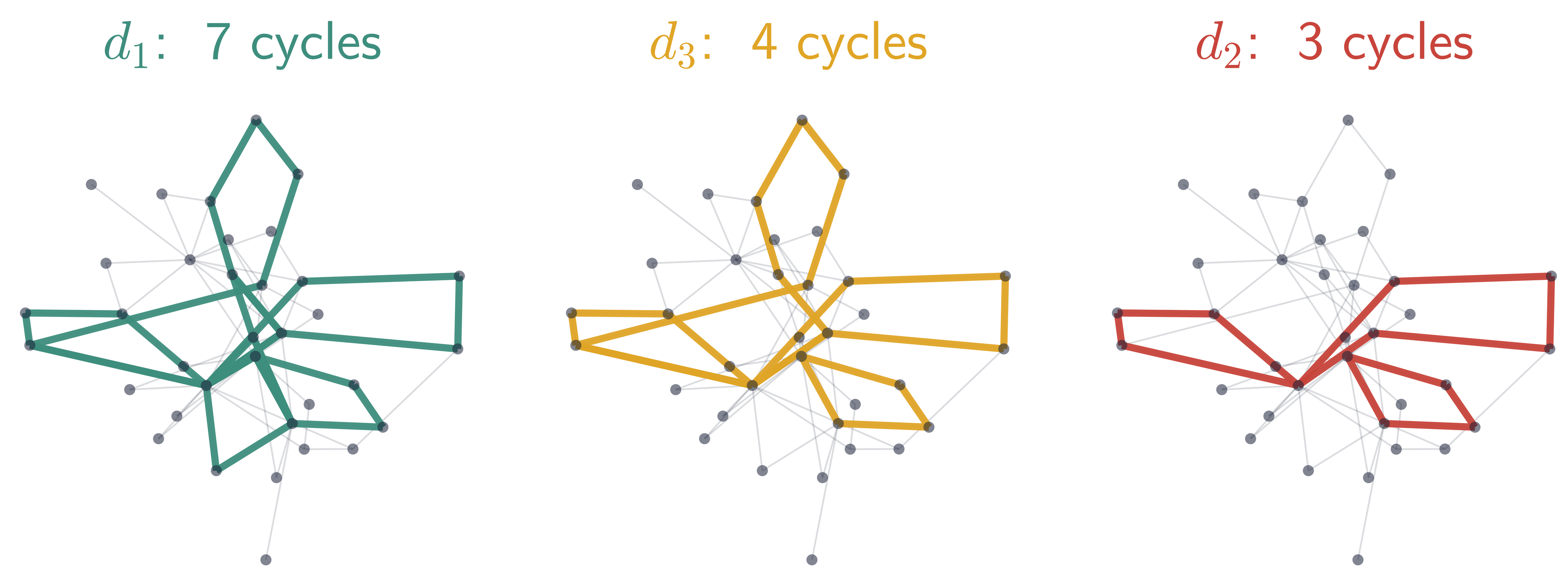
Project page

The big idea

Distance is a **tunable resolution knob** on musical structure — sweeping $d_1 \to d_3 \to d_2$ distills a piece down to its **core cyclic skeleton**.

- **Three path distances**: d_1 (min edges), d_2 (min weight — the only metric), d_3 (hybrid).
- **Proven ordering** $d_2 \le d_3 \le d_1$ lifts to a **nontrivial barcode injection** $\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1)$ (Heo–Choi–Jung 2025).
- **Validated** on real Korean court music — an exact match to the paper's Table 1.

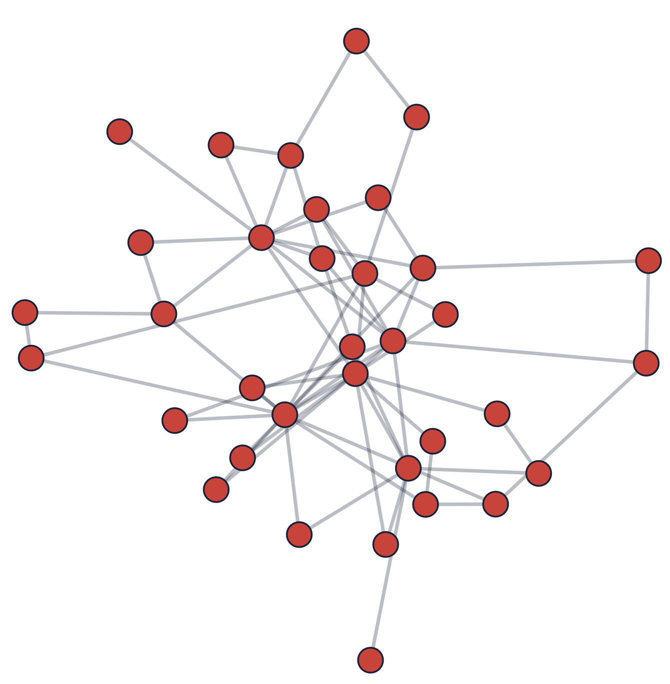
Cycle skeleton extraction



On *Sanghyeondodeuri* (geomungo): as $d_1 \to d_3 \to d_2$, the H_1 barcode thins $|\mathrm{bcd}_1| = \mathbf{7} \to \mathbf{4} \to \mathbf{3}$ — only the robust core loops survive.

From music to a graph

Each **node** = (pitch, duration); an **edge** joins consecutive notes; the **weight** is 1/(co-occurrence count) — frequent neighbors are closer. Rests are skipped.

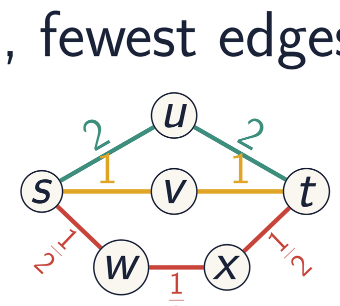


The music network of *Sanghyeondodeuri* (geomungo).

Three path distances

Path P: **hops** $\ell(P)$, **weight** $\omega(P) = \sum_{e\in P} W(e)$; $h = \min_P \ell(P)$.

- d_1 $d_1 = \omega(P),\ \ell(P) = h$ — a **fewest-edge** path.
- d_2 $d_2 = \min_P \omega(P)$ — least weight; *the only metric*.
- d_3 $d_3 = \min\{\omega(P) : \ell(P) = h\}$ — hybrid (least weight, fewest edges).



$s \to t$: d_1=4 (fewest edges) > d_3=2 (fewest edges, least wt.) > d_2=1.5 (least wt.).

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

A universal ordering

For *every* pair of nodes,

$$d_2(v,w) \le d_3(v,w) \le d_1(v,w).$$

This carries over to one-dimensional persistent homology.

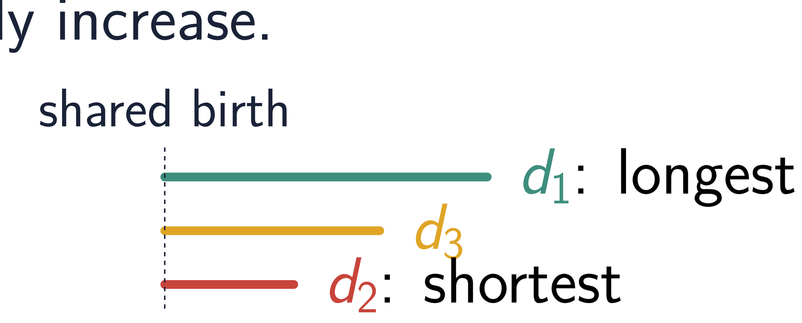
Barcode injection

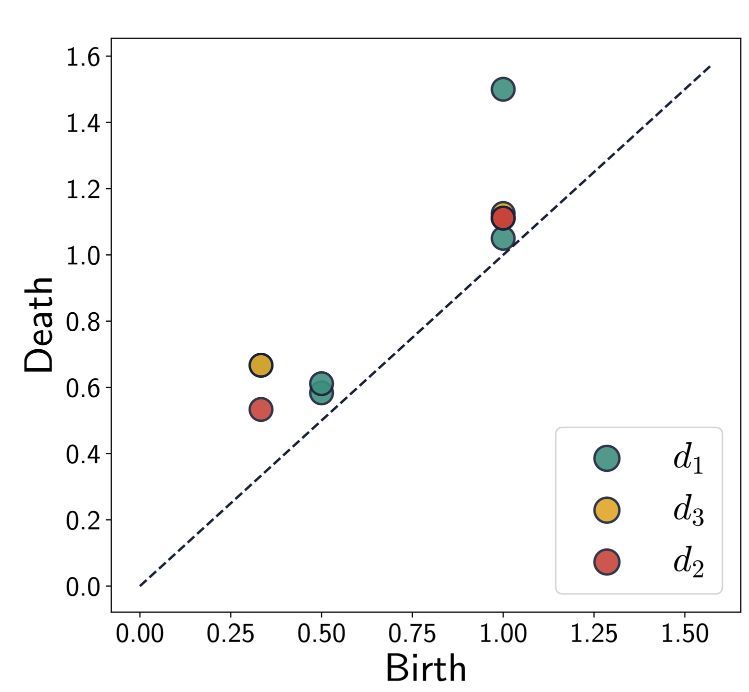
The birth edges nest, $B_2 \subseteq B_3 \subseteq B_1$, giving injections

$$\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1).$$

A **nontrivial theorem** for path-representable distances (**Heo–Choi–Jung 2025**, arXiv:2501.03553).

Each cycle keeps its **birth**; its **lifespan grows** $d_2 \le d_3 \le d_1$, and the number of cycles can only increase.



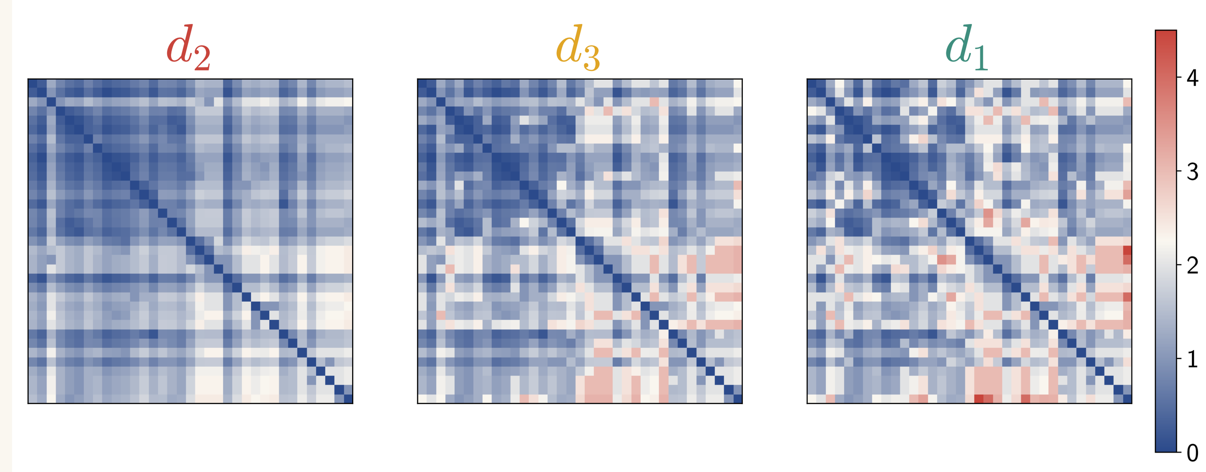


On real music: d_1 points lie above d_2 — longer lifespans, shared births.

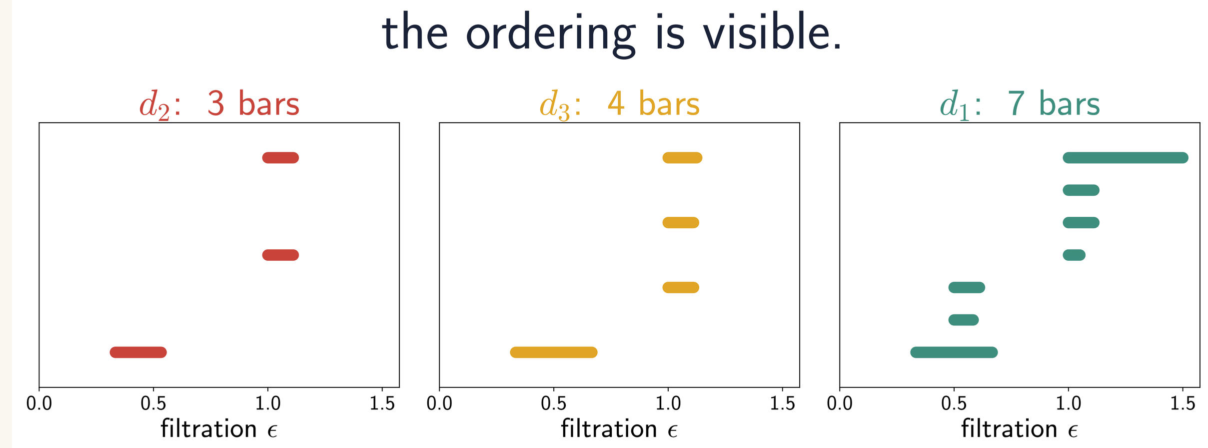
Why it matters

The same piece yields *different but nested* topological readings. d_2 exposes the robust core; d_1 adds finer, peripheral loops. The changes are **systematic, not random**.

Evidence on real music



Distance matrices: d_2 (cool/smooth) → d_1 (hot) the ordering is visible.



H_1 barcodes: $|\mathrm{bcd}_1(d_1)|, |\mathrm{bcd}_1(d_3)|, |\mathrm{bcd}_1(d_2)| = \mathbf{7}, \mathbf{4}, \mathbf{3}$; lifespans grow, births shared.

Distilling the core

Ranking d_1 cycles by how far down they survive ($\to d_3 \to d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.

Interactive explorer & audio demos on the project page.

Key refs: **Heo–Choi–Jung (2025) arXiv:2501.03553** (barcode injection); Tran–Park–Jung (2023) *J. Math. Music*; Sethares–Budney (2014).

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This work: *J. Math. Music* 2025 (arXiv:2506.13595)